



LUCID

Clarity Through the Chaos.

Project Technical Handbook

Architecture & Implementation

VERSION

1.0

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Production Build

Live Application • [GitHub Repository](#)

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1. Executive Summary

LUCID is an AI-powered cybersecurity decision-support platform that turns confusing security evidence into actionable guidance. It adapts the same submitted evidence to two audiences: ShieldMe for everyday users and TIQ for technical analysts.

The validated production state is Cloud Run revision **lucid-00013-xwc**. The primary controlled benchmark achieved **68/74 strict PASS = 91.9%**, with **6 PARTIAL** and **0 FAIL**. ShieldMe achieved **74/74 successful outcomes**. File and Wireshark validation are reported separately to avoid inflating one broad accuracy score.

Important: LUCID is a decision-support tool, not an absolute security guarantee, malware sandbox, penetration test, compliance certification, or replacement for qualified security judgment.

2. Problem, Purpose, and Unique Value

2.1 Problem Solved

Cybersecurity evidence is often fragmented, urgent, and hard to interpret. Everyday users may see suspicious emails, login alerts, payment requests, popups, attachments, or screenshots without knowing whether they are safe. Analysts need structured evidence, severity, indicators, and ATT&CK; context, but manual triage takes time and can be inconsistent.

2.2 Negative Impact if Unsolved

Users may ignore real threats or panic over benign messages. Fraud, credential theft, malware execution, or account compromise may go uncontained. Security analysts may spend extra time translating fragmented evidence into triage decisions, slowing response and weakening auditability.

2.3 What Makes LUCID Different

LUCID combines dual-mode user experience, safe file handling, multimodal evidence support, deterministic evidence-first safeguards, incident correlation, analyst notes, status workflow, Risk Register support, and validation across core, generalization, adversarial, file, and Wireshark scenarios.

3. System Architecture

The architecture separates user interface, authentication, backend validation, safe preprocessing, AI inference, deterministic evidence consistency, and tenant-scoped persistence. User identity comes from Firebase token verification and is not trusted from client-submitted data.

3.1 Architecture Flowchart

LUCID System Architecture

From input to insight. Secure. Scalable. Built for a safer tomorrow.

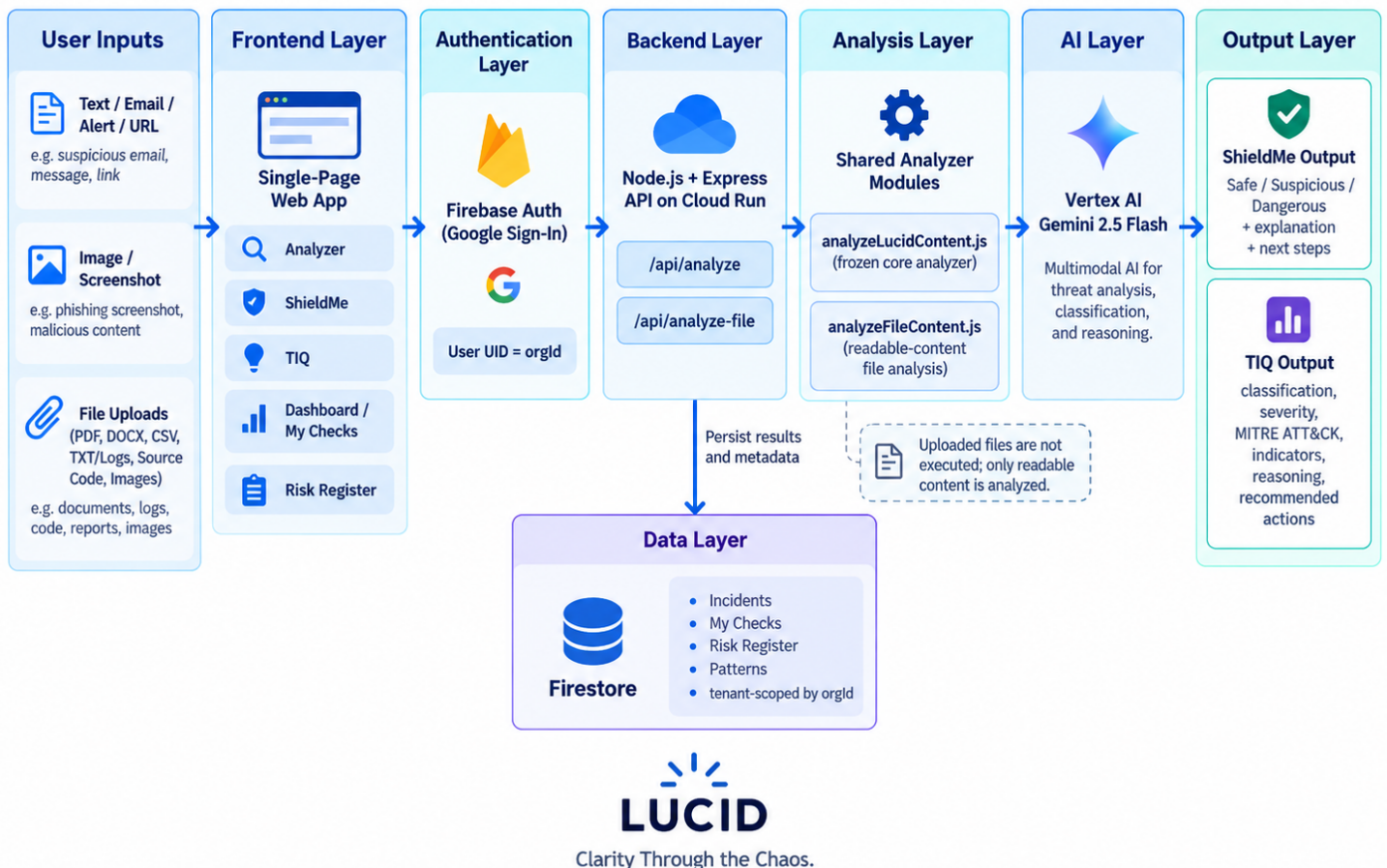


Figure 1 - System architecture. Evidence moves through the frontend, Firebase Auth, Cloud Run backend, shared analyzer modules, Vertex AI Gemini, Firestore, and ShieldMe/TIQ outputs.

3.2 End-to-End Request Flowchart

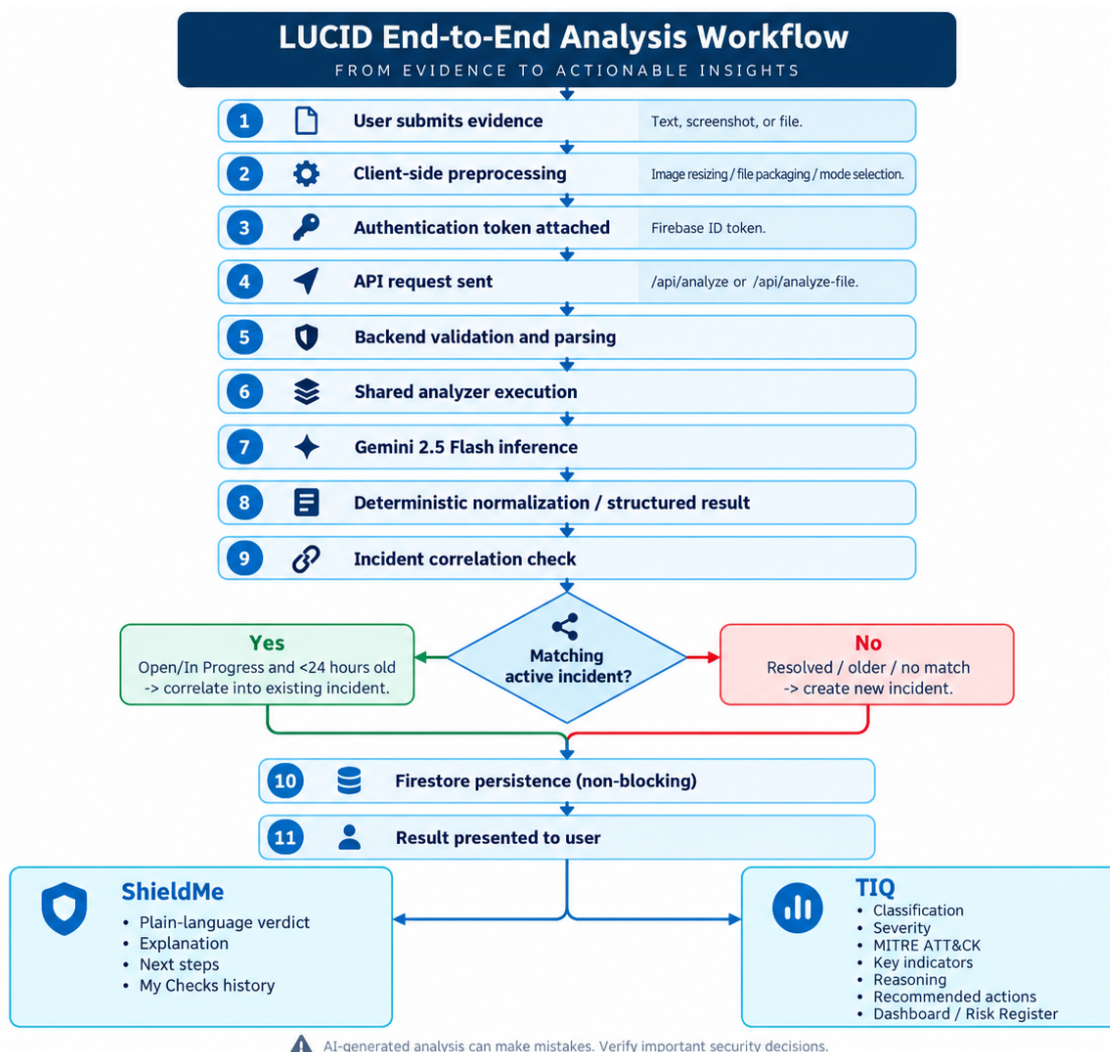


Figure 2 - End-to-end workflow from submitted evidence to structured result, incident correlation, and user-facing output.

4. Core AI Analysis Pipeline

The shared production analyzer is the single source of truth for prompt construction, JSON schema handling, response validation, and evidence consistency. Production API routes and benchmark runners use the same analyzer path, reducing evaluation-to-production drift.

Component	Current behavior
Analyzer module	lib/analyzeLucidContent.js is frozen unless a full regression cycle is intentionally started.
File analyzer	lib/analyzeFileContent.js safely extracts readable content and aggregates file evidence.
Model	Vertex AI Gemini 2.5 Flash in us-central1.
Output schemas	ShieldMe and TIQ use separate structured JSON contracts.
Evidence-first policy	Deterministic safeguards prevent unsupported escalation or unsafe de-escalation.

5. User Modes: ShieldMe and TIQ

Mode	Audience	Output
ShieldMe	Everyday users	Safe/Suspicious/Dangerous verdict, plain-language explanation, and next steps.
TIQ	Analysts and reviewers	Classification, severity, MITRE ATT&CK mapping, indicators, reasoning, unknowns, conf

ShieldMe hides analyst dashboard complexity and focuses on clear action. TIQ exposes Dashboard, incident review, Risk Register, and technical fields needed for triage and response documentation.

6. File and Multimodal Analysis

LUCID supports text and image analysis plus a safe file pipeline for readable files. Uploaded files are treated as evidence/data and are not executed. Supported MVP inputs include TXT/MD/LOG, CSV, source code, PDF, DOCX, and image/Wireshark screenshots.

6.1 File Analysis Pipeline Flowchart

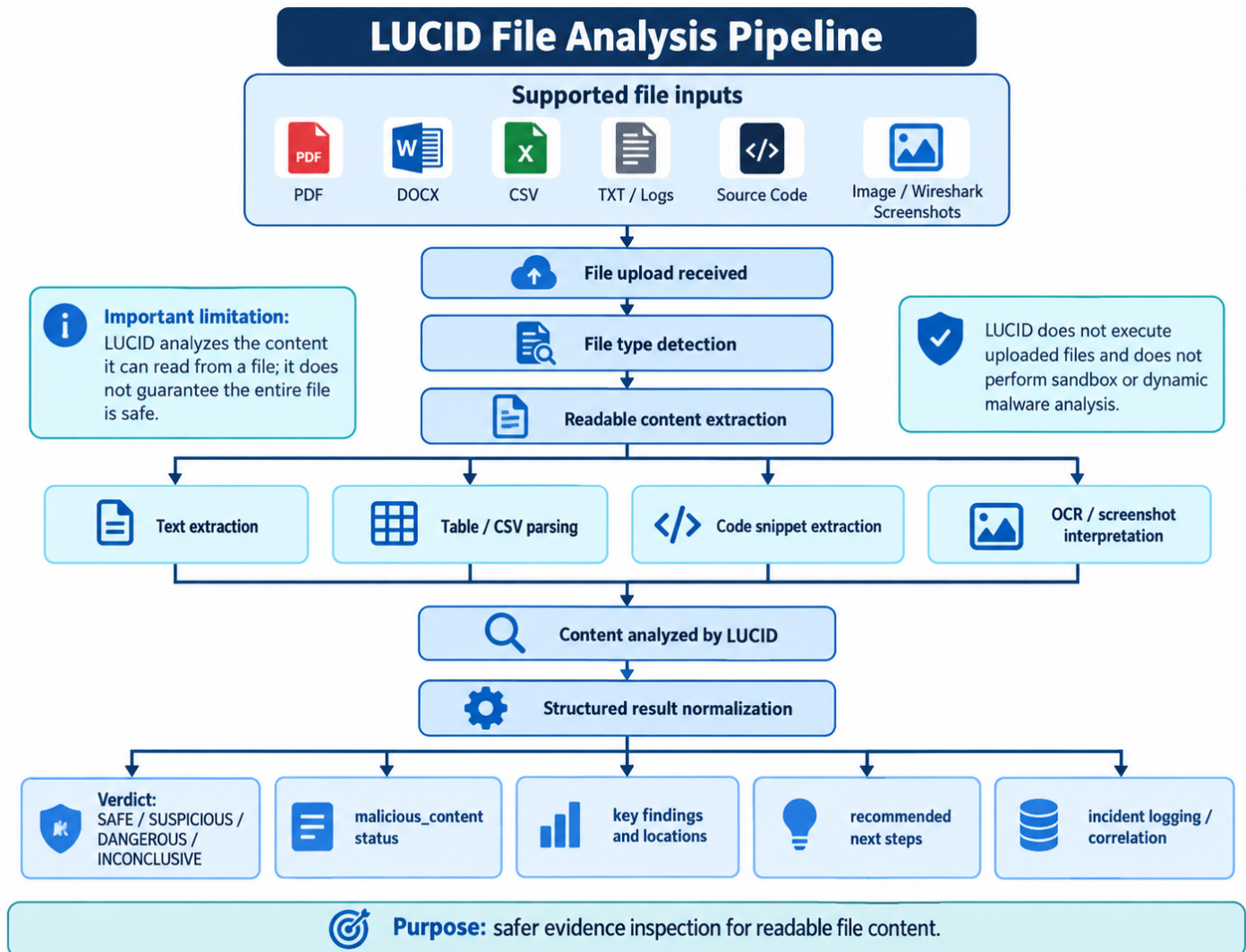


Figure 3 - File analysis pipeline. Uploaded files are validated, parsed for readable content, analyzed, normalized, and returned with verdict, findings, and limitations.

Documentation-aware file analysis update

LUCID now detects software/project documentation as benign content using content-only signals. Markdown, HTML snippets, architecture text, deployment instructions, and security terminology inside documentation are not treated as malicious unless direct malicious behavior is present. Raw phishing and malware evidence detection is preserved. Regression: 19/19 PASS, 0 FAIL.

File results can include **SAFE**, **SUSPICIOUS**, **DANGEROUS**, or **INCONCLUSIVE**. The file pipeline also reports **malicious_content** as detected, not_detected, or inconclusive. Unknown, unreadable, unsupported, or ambiguous files should not be marked Safe by default.

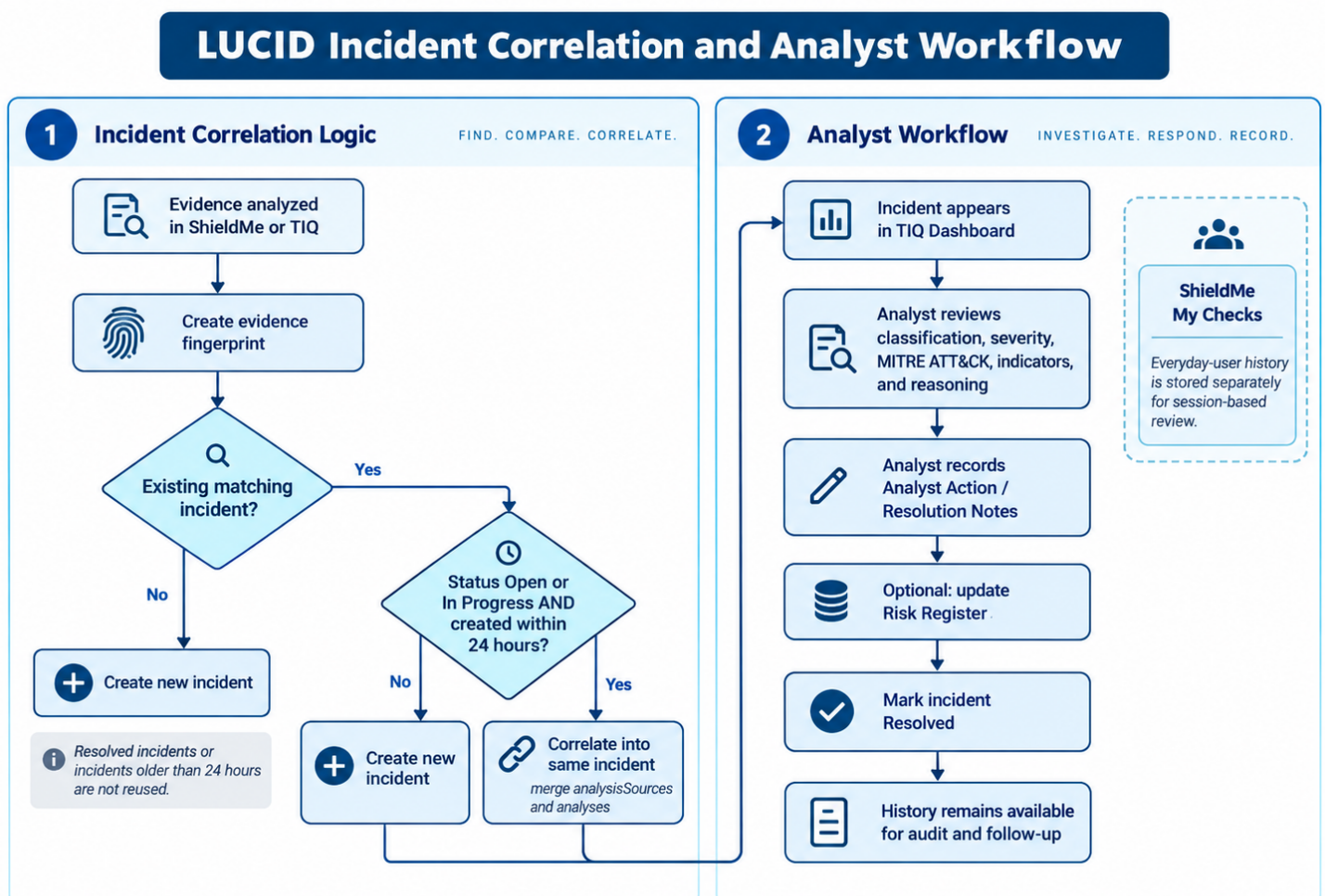
Required limitation: LUCID analyzed the content it could read from this file. This does not guarantee that the entire file is safe.

For Wireshark screenshots, LUCID distinguishes reconnaissance from compromise. Scan-only evidence should remain Active Scanning/Medium in TIQ and Suspicious in ShieldMe unless beaconing, exploitation, or exfiltration is actually visible.

7. Incident Correlation and Dashboard Workflow

LUCID is not only a one-off answer generator. Threat-relevant analysis can become an incident, and the same evidence analyzed in ShieldMe and TIQ can be correlated into one active incident when conservative matching rules are satisfied.

7.1 Incident Correlation and Analyst Workflow



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PEOPLE | INTELLIGENCE | SAFER TOMORROW

Figure 4 - Incident correlation and analyst workflow. Matching uses evidence fingerprints, active status, and a 24-hour window; resolved or older incidents are not reused.

7.2 Dashboard Enhancements, Risk Register, and Pattern Detection

Enhancement	Purpose
Friendly Incident IDs	Human-readable incident tracking in the Dashboard and modal.
Source column	Shows ShieldMe, TIQ, or ShieldMe + TIQ when the same evidence is correlated.
Quick filters	All, Open Only, and Critical Only filters support fast triage.
Incident modal	Shows evidence summary, analysis details, status, and analyst notes.
Resolution notes	Resolved incidents require notes for auditability.
Risk Register	Supports controlled edit/delete workflow for risk records.
Pattern Detection	Groups repeated or related patterns for analyst review.

8. Security, Authentication, and Data Isolation

Protected API routes require a Firebase Bearer ID token. Firebase Admin verifies the token, and the decoded UID becomes the server-derived orgId. Firestore queries and mutations are scoped to that orgId. Client-supplied tenant identity is not trusted.

Control	Current behavior
Authentication	Firebase Google Sign-In with backend token verification.
Tenant isolation	orgId is derived server-side from the verified token.
Mutation safety	PATCH routes use allow-listed fields and ownership checks.
Image safety	Sharp strips metadata and resizes/re-encodes images.
File safety	Uploaded files are parsed as data and not executed.
CSV export	Formula-injection prefixes are neutralized where applicable.

9. Validation and Accuracy Results

The primary controlled benchmark contains 74 cases across Core Regression, Generalization, and Adversarial suites. File and Wireshark validations are reported separately so the project does not inflate one broad accuracy number.

Validation area	Result
Core Regression	24/26 strict PASS = 92.3%; 2 PARTIAL; 0 FAIL
Generalization	18/18 strict PASS = 100%; 0 FAIL
Adversarial	26/30 strict PASS = 86.7%; 4 PARTIAL; 0 FAIL
Combined primary benchmark	68/74 strict PASS = 91.9%; 6 PARTIAL; 0 FAIL
ShieldMe primary benchmark	74/74 successful outcomes
File + doc/threat validation	File smoke 19/19; doc/threat 19/19 PASS, 0 FAIL
Extended file validation	17/17 PASS
Wireshark screenshot validation	16/16 PASS

Interpretation: The 91.9% result is a controlled project-validation score. It is not a universal real-world detection guarantee.

10. Deployment and Operations

Operational item	Current state
Cloud Run service	lucid
Region	us-central1
Validated revision	lucid-00013-xwc
Production URL	https://lucid-dmhlvl2qqa-uc.a.run.app
Google Cloud project	project-c48afffb-501b-4711-a6d
Final analyzer hash	009ad4427809bce6b8eaf9f932714587d7bb9e4f425a2d08684c06a18c2b5208
Health endpoint	No /health route. Cannot GET /health is expected; use the root URL for smoke checks.

11. Limitations, Technical Debt, and Future Work

Area	Current limitation / future direction
AI judgment	Decision support only; important security actions require verification.
File analysis	No dynamic malware sandbox; unreadable, encrypted, packed, or hidden content may remain unseen.
SDK lifecycle	Vertex AI SDK path should migrate to @google/genai with full regression testing.
Threat intelligence	Future IOC, URL reputation, WHOIS/domain age, and enrichment integration.
Enterprise integrations	Future SIEM/SOAR integrations such as Google SecOps, Splunk, Microsoft Sentinel, QRadar.
Enterprise controls	Future RBAC, audit logs, SSO/SAML, and integration-secret management.

12. Reviewer Walkthrough

A reviewer can understand the project by opening the live application, running the same evidence through ShieldMe and TIQ, viewing incident correlation in the Dashboard, analyzing a supported file, checking Wireshark scan behavior, reviewing My Checks and Risk Register, and then comparing the observed behavior with the Accuracy Report and Test Cases.

Appendix A. Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript single-page app
Backend	Node.js / Express
AI	Vertex AI Gemini 2.5 Flash
Authentication	Firebase Authentication + Firebase Admin verification
Database	Cloud Firestore
Deployment	Cloud Run, Cloud Build, Artifact Registry
File/Image	multer, sharp, pdf-parse, mammoth, csv-parse
Validation	Node.js benchmark and evaluator scripts

Appendix B. API Reference

Endpoint family	Purpose
/api/analyze	Primary ShieldMe/TIQ text/image analysis endpoint.
/api/analyze-file	File upload analysis endpoint for readable supported files.
/api/incidents	Tenant-scoped incident listing and detail workflow.
/api/incidents/:id	Controlled incident status/notes update.
/api/my-checks	ShieldMe history for checks that may need attention.
/api/risks and /api/risks/:id	Risk Register list/create/update/delete workflow with ownership checks.

Appendix C. Evidence Log

Evidence item	Verified release state
Production application	https://lucid-dmhlvl2qqa-uc.a.run.app
Cloud Run service / region	lucid / us-central1
Validated revision	lucid-00013-xwc
Final analyzer hash	009ad4427809bce6b8eaf9f932714587d7bb9e4f425a2d08684c06a18c2b5208
Primary benchmark	68/74 strict PASS = 91.9%; 6 PARTIAL; 0 FAIL
ShieldMe	74/74 successful outcomes
Separate validations	File 19/19; extended file 17/17; Wireshark 16/16; doc/threat 19/19
Release interpretation	Controlled validation only; not universal accuracy or compliance certification.